



GM UNIVERSITY

IEEE STUDENT BRANCH

Lecture Series Nov-Dec 2025



OUR SPEAKER

Mr. Pruthvisagar G P

Corpnce Technologies Lead Machine Learning
Instructor

Technical Talk: Linear Algebra for Machine Learning



Patrons

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Dean-Research

Dr. C S Ramesh
Research-Advisor

Faculty Co-ordinator : Dr. Nasreen Taj M B

Date: 21\11\2025 Venue : Civil Seminar Hall Time: 11 AM to 1PM



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Post Box no-4, PB Road, Davangere-577006

FACULTY OF ENGINEERING AND TECHNOLOGY

School of Computer Science and Technology

Event Report: Technical Talk on Linear Algebra for Machine Learning

The **IEEE Student Branch of GM University**, in association with the **Department of Artificial Intelligence and Machine Learning (AIML)**, successfully conducted a Technical Talk under the **Lecture Series (Nov–Dec 2025)**. The session aimed to strengthen the mathematical foundation required for understanding core Machine Learning concepts.

Speaker Details

Mr. Pruthvisagar G. P

Lead Machine Learning Instructor, Corpncce Technologies

AIML Alumni, GM University (2023–24 outgoing batch)

Event Details

 **Date:** 21 November 2025

 **Time:** 11:00 AM – 1:00 PM

 **Venue:** Civil Seminar Hall, GM University

Session Highlights

The session focused on the importance of **Linear Algebra** as a foundational pillar for understanding various Machine Learning algorithms and deep learning techniques. Topics covered included:

- Vector spaces and matrix operations
- Eigenvalues and eigenvectors in ML
- Linear transformations
- Use of matrices in optimization and learning algorithms
- Real-time applications of linear algebra in AI and ML models

Mr. Pruthvisagar explained each concept with simple examples, making it easier for students to connect mathematical theory with practical machine learning applications. Students actively interacted with the speaker and gained clarity on several fundamental doubts.



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Outcome

The session was highly insightful and extremely beneficial for students pursuing AI and ML. It provided clarity on key mathematical concepts and encouraged students to strengthen their theoretical understanding, which is crucial for advanced ML topics.

The IEEE Student Branch and the AIML Department look forward to organizing more such knowledge-enriching sessions.





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Pruthvisagar G P

MACHINE LEARNING INSTRUCTOR

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Summary

Machine Learning Instructor with a strong foundation in AI, Data Science, and Applied Mathematics. Passionate about simplifying complex ML concepts by connecting mathematical intuition with real-world applications. Experienced in designing structured, math-focused ML courses that bridge the gap between theory and practice through intuitive visualisations and interactive demonstrations using graphing calculators. Skilled at guiding learners through core topics such as Linear Algebra, Statistics, and Deep Learning, helping them build a deep conceptual and graphical understanding of Machine Learning

Experience

Corpnce Technologies | Lead Machine Learning Instructor **2024-Present**

- **Curriculum development:** Architected a progressive syllabus that maps mathematical theory → algorithm derivation → practical implementation, enabling learners to move from intuition to code in 6–8 weeks.
- **Hands-on pedagogy:** Built reproducible lab notebooks and graded projects where students implement linear regression, PCA, gradient descent, Bayesian inference and regularization techniques using real datasets.
- **Outcomes & mentorship:** Ran weekly office hours and project reviews; guided capstone teams from problem scoping to production-level evaluation metrics and demo presentations.

Projects

• BPE Tokenizer

I trained a small BPE tokenizer on a custom text dataset to see how vocab size affects sequence length and Transformer training. Tried 5k, 8k and 16k vocab sizes, compared token splits, and plugged the tokenizer into a small encoder model to observe how training loss and memory use changed.

• LLM configs eval

I ran a set of experiments on a small Transformer model by changing only core architectural settings: number of heads, embedding size, depth, dropout and learning rate schedules. Tracked how these changes affected stability, training speed and validation loss. The project helped me understand which parts of the architecture impact performance the most.

Education

GM Institute of technology(VTU), BE in AI ML

2020-2024